A weighted q-Mehler limit for selected preparations

Global spectral control and the first Rindler-time correction

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Abstract

We prove the compact-time preparation correction predicted in the preceding source-response note. For fixed positive integer Mehler dimension d, finite product shifts and the theta modular transform give a globally dominated center-energy/gap density. Its first weighted expansion yields an $O(\varepsilon^2)$ remainder, including any fixed number of real-time derivatives. For the actual curvature channel d=2, the tracial and single-selected correlations have equal and opposite first corrections to $\operatorname{sech}^4 u$. Their equal statistical mixture therefore removes this correction. The result concerns sequential double-scaling and small-deformation limits, not a particle-mass identification or finite-G dynamics.

1 Objects and result

Put $q = e^{-\varepsilon}$, $\varepsilon = 4\lambda > 0$, and let μ_q be the unit-variance q-Hermite measure with $e_\theta = 2\cos\theta/\sqrt{1-q}$, $0 < \theta < \pi$. For a fixed positive integer d, let $\nu_{q,d}$ be its normalized two-energy Mehler measure with parameter $r = q^d$ [1]. Define the gap $\omega = (e_\theta - e_\phi)/\sqrt{\varepsilon}$ and

$$C_{d,\varepsilon}(u) = \int e^{iu\omega}\, d\nu_{q,d},$$
$$\widetilde{C}_{d,\varepsilon}(u) = \int [1 - \sqrt{q} + \sqrt{q}\, e_\theta^2] e^{iu\omega}\, d\nu_{q,d}.$$

Both are normalized positive spectral characteristic functions. The second uses a positive energy tilt; for d=2 it is the already proved single-selected curvature correlation [5, 6]. For other d no new identification with a microscopic selected field is assumed.

Theorem 1. *For fixed d, $S < \infty$ and nonnegative integer k, as $\varepsilon \downarrow 0$,*

$$\begin{aligned} C_{d,\varepsilon} &= g_d - \tfrac{1}{2}\varepsilon R_d + O_{C^k([-S,S])}(\varepsilon^2), \\ \widetilde{C}_{d,\varepsilon} &= g_d + \tfrac{1}{2}\varepsilon R_d + O_{C^k([-S,S])}(\varepsilon^2), \\ g_d(u) &= \operatorname{sech}^{2d} u, \qquad R_d(u) = \tfrac{d}{2} g_d(u)[u\tanh u - (2d+1)\tanh^2 u]. \end{aligned} \tag{1}$$

For complex DSSYK, first take the half-filled fixed-λ limit with the original common disorder and standard CAR. Then $K = H/(2J_s) = e/\sqrt{\varepsilon}$ and $u = 2J_s t$ use string time, $\hbar = 1$. The theorem concerns this limiting kernel; no uniform finite-N rate or joint N, λ limit is asserted.

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2 Exact product and modular representation

Write $Q_\varepsilon = (q;q)_\infty$ and $A_\varepsilon(x) = |(e^{ix};q)_\infty|^2$. The known angular measure and Poisson kernel give

$$\frac{d\nu_{q,d}}{d\theta\, d\phi} = \frac{Q_\varepsilon^2}{4\pi^2} \frac{A_\varepsilon(2\theta) A_\varepsilon(2\phi)(q^{2d};q)_\infty}{|(q^d e^{i(\theta+\phi)}, q^d e^{i(\theta-\phi)};q)_\infty|^2}. \tag{2}$$

The multiple-argument Pochhammer denotes a product. Its finite shift is exact: $|(q^d e^{ix};q)_\infty|^2 = A_\varepsilon(x)/\prod_{j=0}^{d-1}(1-2q^j\cos x + q^{2j})$.

The theta product and modular transform [2] imply

$$\begin{aligned} A_\varepsilon(x) &= \frac{2e^{\varepsilon/8}\sin(x/2)}{Q_\varepsilon}\sqrt{\frac{2\pi}{\varepsilon}}\, T_\varepsilon(x), \\ T_\varepsilon(x) &= \sum_{n\in\mathbb{Z}} (-1)^n e^{-(x-\pi-2\pi n)^2/(2\varepsilon)}, \\ Q_\varepsilon &= \sqrt{\frac{2\pi}{\varepsilon}} e^{-\pi^2/(6\varepsilon)+\varepsilon/24} P_\varepsilon, \quad P_\varepsilon = (e^{-4\pi^2/\varepsilon}; e^{-4\pi^2/\varepsilon})_\infty. \end{aligned} \tag{3}$$

Use center/gap coordinates

$$a = \theta + \phi - \pi = \sqrt{\varepsilon}\, y, \quad \delta = \theta - \phi = \varepsilon z, \quad d\theta d\phi = \tfrac{1}{2}\varepsilon^{3/2} dy\, dz.$$

The domain is $\mathcal{D}_\varepsilon = \{\sqrt{\varepsilon}|y| + \varepsilon|z| < \pi\}$. Let $p_{d,\varepsilon}$ be the transformed density, extended by zero outside. It is even separately in y and z. Below take $z \geq 0$ and extend evenly. Set $S_\theta = \pi + a$, $D_m = \prod_{j=1}^m (1-e^{-j\varepsilon})$, and

$$B_j^+ = 2e^{-j\varepsilon}(\cosh(j\varepsilon) + \cos a), \quad B_j^- = 2e^{-j\varepsilon}(\cosh(j\varepsilon) - \cos\delta).$$

Equations (2)–(3) give exactly

$$\begin{aligned} p_{d,\varepsilon}(y,z) = \frac{(2\pi)^{3/2} P_\varepsilon^3}{\pi^2 D_{2d-1}} e^{\varepsilon/8 - y^2/2 - \varepsilon z^2/2} \sin\theta \sin\phi \sin(S_\theta/2) \\ \times \frac{\sin(\varepsilon z/2)}{\sinh(\pi z)} \prod_{j=1}^{d-1} B_j^+ B_j^-\, \mathcal{E}_\varepsilon(y,z). \end{aligned} \tag{4}$$

The diagonal quotient has limit $\varepsilon/(2\pi)$. The factor $\mathcal{E}_\varepsilon$ is specified next, including all reflected images.

3 Images, edges and a global integrable bound

For $0 \le x \le 2\pi$ set $b(x) = \min(x, 2\pi - x)$, $F_\varepsilon(x) = 1 - e^{-2\pi b(x)/\varepsilon}$. Pairing reflected terms in (3) gives

$$T_\varepsilon(x) = e^{-(x-\pi)^2/(2\varepsilon)} F_\varepsilon(x) Q_\varepsilon^{\mathrm{im}}(x),$$

where the endpoint quotient is defined by its limit and

$$Q_\varepsilon^{\mathrm{im}}(x) = \sum_{n\ge 0} (-1)^n e^{-2\pi n((n+1)\pi - b(x))/\varepsilon} \sum_{j=0}^{2n} e^{-2\pi b(x) j/\varepsilon}. \tag{5}$$

For $0 < \varepsilon \le 1$ this yields the uniform relative bound

$$|Q_\varepsilon^{\mathrm{im}}(x) - 1| \le \eta_\varepsilon := \sum_{n\ge 1} (2n+1) e^{-2\pi^2 n^2/\varepsilon} \le 4 e^{-2\pi^2/\varepsilon} < \tfrac{1}{2}. \tag{6}$$

The termwise bound uses $b \le \pi$; the final geometric estimate follows from $n^2 \ge n$ and $e^{-2\pi^2} < 10^{-8}$. Thus the image factor in (4) is

$$\mathcal{E}_\varepsilon = \frac{F_\varepsilon(2\theta) F_\varepsilon(2\phi)}{F_\varepsilon(S_\theta)} \frac{Q_\varepsilon^{\mathrm{im}}(2\theta) Q_\varepsilon^{\mathrm{im}}(2\phi)}{Q_\varepsilon^{\mathrm{im}}(S_\theta) Q_\varepsilon^{\mathrm{im}}(\delta)}. \tag{7}$$

The factor $F_\varepsilon(\delta)$ was absorbed into $2\sinh(\pi z)$ in deriving (4). This is essential near $z = 0$.

Define a positive normalized gap density and the standard Gaussian:

$$f_d(z) = \frac{2^{2d-1}}{(2d-1)!} \frac{z \prod_{j=1}^{d-1} (j^2 + z^2)}{\sinh \pi z}, \qquad \varphi(y) = \frac{e^{-y^2/2}}{\sqrt{2\pi}}. \tag{8}$$

The removable value at zero is understood. On the whole domain,

$$0 \le p_{d,\varepsilon}(y, z) \le C_d \varphi(y) f_d(z), \qquad 0 < \varepsilon \le 1. \tag{9}$$

Indeed $D_{2d-1} \ge c_d \varepsilon^{2d-1}$, $B_j^+ \le C_j$, $B_j^- \le C_j \varepsilon^2 (j^2 + z^2)$, and $|\sin(\varepsilon z/2)| \le \varepsilon |z|/2$. The only apparent edge singularity is harmless: $\sin(S_\theta/2)/F_\varepsilon(S_\theta) \le 2$. For $b \ge \varepsilon$ use $F \ge 1 - e^{-2\pi}$; for $b \le \varepsilon$ use concavity, $F \ge (1 - e^{-2\pi}) b/\varepsilon$ and $\sin(S_\theta/2) \le b/2$. All other image factors are bounded by (6). This proves (9), including arbitrarily small spectral-edge layers.

4 A weighted density expansion with a controlled remainder

The first relative density coefficient is

$$A_d(y,z) = \frac{4d+1}{8} - \frac{2d+1}{8}y^2 - \frac{1}{2}z^2. \tag{10}$$

For every fixed polynomial weight w in $|y|, |z|$,

$$\int w\ |p_{d,\varepsilon} - \varphi f_d(1+\varepsilon A_d)|\, dy\, dz = O_{d,w}(\varepsilon^2). \tag{11}$$

Here and below weights may be replaced by nonnegative sums of monomials.

To prove it, take $L = \log(1/\varepsilon)$ and the central rectangle $|y| \le 4\sqrt{L}$, $|z| \le 8L/\pi$, for sufficiently small eps. It lies in $\mathcal{D}_\varepsilon$. There $b(2\theta), b(2\phi), b(S_\theta) \ge \pi/2$, so (7) differs from one by $O(e^{-\pi^2/\varepsilon})$, and $P_\varepsilon = 1 + O(e^{-4\pi^2/\varepsilon})$. The remaining ratio $p/(\varphi f_d)$ is a finite product of regular elementary factors. Their first coefficients are

Factor	coefficient of eps in its normalized form
$e^{\varepsilon/8 - \varepsilon z^2/2}$	$1/8 - z^2/2$
$\sin\theta \sin\phi$	$-y^2/4$
$\sin(S_\theta/2)$	$-y^2/8$
$\mathrm{sinc}(\varepsilon z/2)$	0
$B_j^+/4$	$-j - y^2/4$
$B_j^-/[\varepsilon^2(j^2+z^2)]$	$-j$
$(2d-1)!\varepsilon^{2d-1}/D_{2d-1}$	$d(2d-1)/2$

The j rows range from 1 to $d-1$. Summing gives (10). Taylor's formula for sine, cosine and exponentials gives a remainder bounded by $C_d\varepsilon^2 P_d(|y|,|z|)$ on this rectangle, for a fixed polynomial P_d. Denominators j^2+z^2 are bounded below by j^2, and the normalized factors $D_m/(m!\varepsilon^m)$ stay away from zero. Exponential remainder factors are bounded because $\varepsilon(L+L^2) \to 0$. Integrating against $w\varphi f_d$ therefore gives $O(\varepsilon^2)$ with no logarithmic loss.

Outside the rectangle, (9) bounds the true density. Both Gaussian and gap tails are $O(\varepsilon^8)$ times powers of L, also with any fixed polynomial weight. The approximation in (11) obeys the same tail estimate with extra polynomial factors. Such tails are $o(\varepsilon^2)$. This proves (11); the argument does not discard an uncontrolled spectral edge.

5 Energy insertion, phase and Gaussian cancellation

Because the density and gap are even in y, replace $e_\theta^2 - 1$ by its exact y-symmetrization

$$W_\varepsilon(y,z) = \frac{2[1-\cos(\sqrt{\varepsilon}y)\cos(\varepsilon z)]}{1-e^{-\varepsilon}} - 1 = W_0 + \varepsilon W_1 + O(\varepsilon^2 P),$$
$$W_0 = y^2 - 1, \quad W_1 = z^2 + y^2/2 - y^4/12, \tag{12}$$
$$\omega_\varepsilon(y,z) = \frac{-4\cos(\sqrt{\varepsilon}y/2)\sin(\varepsilon z/2)}{\sqrt{\varepsilon(1-e^{-\varepsilon})}} = -2z[1+\varepsilon(1/4 - y^2/8)] + O(\varepsilon^2 P). \tag{13}$$

The remainder polynomials are independent of eps for $0 < \varepsilon \le \varepsilon_0$. Real trigonometric Taylor bounds prove these estimates on the whole domain, with polynomial growth allowed. Also $|W_\varepsilon| \le C(1+y^2+z^2)$ and $|\omega_\varepsilon| \le C|z|$. Thus (11) applies to their products. For $|u| \le S$,

$$e^{iu\omega_\varepsilon} = e^{-2iuz}[1 - 2iuz\varepsilon(1/4 - y^2/8)] + O_S(\varepsilon^2 P).$$

The same bounds hold after any fixed number of u derivatives, with larger polynomial weights.

The difference of the two correlations is exactly $e^{-\varepsilon/2}\int p_{d,\varepsilon} W_\varepsilon e^{iu\omega_\varepsilon} dy\, dz$. Since $\int \varphi W_0 = 0$, its constant term vanishes. Using $\int \varphi y^2 = 1$ and $\int \varphi y^4 = 3$, its first coefficient is

$$\int \varphi(y)\left\{W_1 + W_0\left[A_d - \tfrac{1}{2} - 2iuz(\tfrac{1}{4} - \tfrac{1}{8}y^2)\right]\right\} dy$$
$$= z^2 - \frac{d}{2} + \frac{iuz}{2}. \tag{14}$$

Every discarded integral is $O_{S,k}(\varepsilon^2)$ by (11) and the polynomial bounds above. The term $iuz/2$ is the phase correction; dropping it would lose the $u\tanh u$ term.

The beta integral and Fourier inversion give the familiar identity

$$\int f_d(z) e^{-2iuz} dz = g_d(u) = \operatorname{sech}^{2d} u. \tag{15}$$

For example $f_d = 2^{2d-1}|\Gamma(d+iz)|^2/[\pi\Gamma(2d)]$; the substitution $s = (1+\tanh u)/2$ proves its Fourier pair. All polynomial moments of f_d are finite. Equation (14) therefore yields

$$\widetilde{C}_{d,\varepsilon} - C_{d,\varepsilon} = \varepsilon[-g_d''/4 - d g_d/2 - u g_d'/4] + O_{C^k}(\varepsilon^2) = \varepsilon R_d + O_{C^k}(\varepsilon^2).$$

This directly proves the formerly conditional preparation response.

6 Trace correction, an equal mixture, and interpretation

The same expansion without the weight W gives

$$C_{d,\varepsilon} = g_d + \varepsilon\left[\frac{d}{4}g_d + \frac{1}{8}g_d'' + \frac{1}{8}ug_d'\right] + O_{C^k}(\varepsilon^2) = g_d - \frac{1}{2}\varepsilon R_d + O_{C^k}(\varepsilon^2).$$

Together with the preceding difference this proves Theorem 1. For the physical d=2 curvature channel,

$$C_1 - C_0 = \varepsilon\,\mathrm{sech}^4 u[u\tanh u - 5\tanh^2 u] + O_{C^k}(\varepsilon^2).$$

The exact source identity of [6] now implies $\partial_\beta^2 G_{\varepsilon,\beta}(u)|_0 = R_2(u) + O_{C^k}(\varepsilon)$. No differentiation of an unproved complex-temperature limit is used.

An immediate constructive consequence is the positive state $\sigma_{\mathrm{mix}} = (\tau+\sigma_1)/2$. On the charged subalgebra its density is $(1+f_q(e))/2$, and

$$\tfrac{1}{2}(C_0 + C_1) = \mathrm{sech}^4 u + O_{C^k([-S,S])}(\varepsilon^2). \tag{16}$$

This is an equal statistical mixture of two already controlled preparations; it is not an identified bulk vacuum. No matching claim for general neutral-containing bilocals follows from (16). Their full selected density retains W. The coefficient changes a state-dependent boost correlation, not an invariant particle mass.

7 Evidence and scope

Fresh checks compare the unshifted Pochhammer kernel against the modular expression in 27 cases, with $d = 1, 2, 3$ and $\varepsilon = 0.4, 0.1, 0.04$, including $z = 0$. Six additional cases approach spectral endpoints and the diagonal. At 70 decimal digits the largest relative differences are $1.36\,10^{-68}$ and $6.83\,10^{-67}$. Eight paired-image terms have a relative omitted bound below $2.10\,10^{-1735}$ at $\varepsilon = 0.4$.

Eighteen local-expansion cases test the density, energy weight and gap; symbolic Gaussian integration verifies (14) for general d. Twelve independent real-axis Fourier integrals at 50/70 digits verify the final response for $d = 1, 2, 3$, $u = 0, 0.5, 1, 2$. The integration cutoff $|z| = 32$ has an explicit exponential tail; the maximum absolute discrepancy is $2.76\,10^{-34}$ and the largest tail bound is $2.83\,10^{-33}$. Quadrature roundoff is not interval-certified.

A separate exact q-polynomial calculation verifies sixteen trace Taylor coefficients through u^{14}. Read-only reuse of the preceding 28 packet values checks the mixture's second-order scaling; no successful old scientific driver is rerun. These diagnostics check representations and coefficients. The global proof, not a small grid, establishes the uniform error order.

8 Continuation and provenance

This resolves the specific differentiated-response gap identified in the previous note, for fixed integer Mehler channels and fixed compact real times. The older note and its conditional wording are preserved as a dated record. The leading Mehler limit, theta identities and thermal saddle are established mathematics and prior SYK results; no priority claim is made for them. The present derivation keeps the selected energy insertion, the phase correction and a global spectral remainder in one normalization.

The next physical task is the mixed e,W source of the full actual bilocal: its neutral terms cannot be represented by this energy tilt alone. Higher-point state response and microscopic spatial or mass generators remain to be built. The order of limits stays finite-N exact projection, fixed-λ double scaling, then small lambda at fixed string time. Finite-G dynamics, growing-time limits and arbitrary selected interlacings are not completed here.

References

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